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Inventor(s)	FURMAN; Wojciech et al.

VISE HOLDING SYSTEM FOR PARALLELS AND/OR WORKPIECES

Abstract

Add-on components for a workpiece holder or vise, including a pair of jaws. A work piece lock element is moveable and lockable along a top surface of one of the pair of jaws. The work piece lock element can be or include a positive lock, such as including a threaded element, a rounded or spherical element, and/or a spring loaded ball. In embodiments, a row of detents is disposed in one of the jaws, and each of the detents is configured to receive the positive lock in one position. An adjustable side extension can be included for laterally securing workpieces in the holder. A pair of clips, each attached to a side of one of the pair of jaws, can be used to secure a parallel against the one of the pair of jaws. Parallel keepers can be used to reduce lateral displacement upon tightening the jaws.

Inventors: **FURMAN; Wojciech (Crystal Lake, IL), BOGACZ; Dominik P. (Crystal Lake, IL)**

Applicant: **VARIABLE OPERATIONS TECHNOLOGIES, INC.** (CRYSTAL LAKE, IL)

Family ID: **96881661**

Assignee: **VARIABLE OPERATIONS TECHNOLOGIES, INC. (CRYSTAL LAKE, IL)**

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] This application claims the benefit of U.S. Provisional Application, Ser. No. 63/560,017, filed on 1 Mar. 2024. The co-pending provisional application is hereby incorporated by reference herein in its entirety and is made a part hereof, including but not limited to those portions which specifically appear hereinafter.

BACKGROUND OF THE INVENTION
[0002] This invention relates generally to vises and similar workpieces holders, and more particularly to components for use in/with vises to improve workpiece holding.
SUMMARY OF THE INVENTION
[0003] A general object of the invention is to provide improved workpiece holding in vises, such as using parallels, and more preferably to workplace vise components or accessories that minimize setup time during a machining process of various parts that need to be held in the vise.
[0004] Embodiments of this invention reduce or eliminate the problem of vise parallels not being held firmly enough when the vise is open to remove the machined part and/or blow away any debris. Current products available on the market generally create a movement risk while repositioning parallel or in chips going under the parallels during the cleaning process. This can cause damage to the machined part or the cutting tool.
[0005] In some embodiments, another object is to free the space under the machined components. Embodiments of the invention provide free space below the machined components to allow an endmill's section closer to the tool holder, which helps maintain the rigidness of the tool and improves the surface finish of the machined component. In addition the space cases cleanliness and chip evacuation.
[0006] The components of this invention provide options for improved holding of workpieces and/or parallels, over current devices.
[0007] Embodiments of this invention include improvements or improved add-on components for a workpiece holder, such as including a pair of jaws. In embodiments. the improvement includes a lock element and/or retainer clip(s) connected to one of the pair of jaws.
[0008] The lock element can be a work piece lock element that is moveable and lockable along a top surface of one of the pair of jaws. The work piece lock element can be locked in place by a positive lock, such as including a threaded element, a rounded or spherical element, and/or a spring loaded ball. As used herein, “positive lock” generally refers to securing components through direct mechanical engagement, preventing relative movement without relying solely on friction, pressure, or torque retention. In embodiments, a row of detents is disposed in one of the jaws, and each of the detents is configured to receive the positive lock. For example, in embodiments the threaded element can push against a ball, or lock the spring loaded ball in a detent.
[0009] In embodiments, the improvement additionally or alternative includes an adjustable side extension for laterally securing workpieces in the holder. The side extension can be adjusted longitudinally and/or laterally.
[0010] In embodiments, the improvement additionally or alternative includes a pair of clips, each attached to a side of one of the pair of jaws and together configured to secure a parallel against the one of the pair of jaws.
[0011] In embodiments, the improvement additionally or alternative includes a parallel keeper including a first keeper element including a female detent and a second keeper element including a

male end that fits within the female detent to reduce lateral displacement upon tightening the jaws.
[0012] Other objects and advantages will be apparent to those skilled in the art from the following detailed description taken in conjunction with the appended claims and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0013] FIGS. **1-6** show an exemplary vise including several components of this invention in combination.
[0014] FIGS. **7-16** show a lock according to embodiments of this invention. FIG. **12** is a sectional view along A-A in FIG. **10**. FIG. **13** is a sectional view along B-B in FIG. **10**. FIG. **14** is a sectional view along C-C in FIG. **10**. FIG. **15** is a sectional view along E-E in FIG. **11**. FIG. **16** is an enlarged view of (F) in FIG. **15**.
[0015] FIG. **17** shows a lock according to embodiments of this invention.
[0016] FIGS. **18-21** show a lock according to embodiments of this invention.
[0017] FIGS. **22** and **23** show opposing adjustable side extensions according to embodiments of this invention.
[0018] FIGS. **24-25** show parallel clips and vise parallels according to embodiments of this invention.
[0019] FIGS. **26-30** show a parallel holder or parallel keeper disposed between parallels, and in combination with the parallel clips, according to embodiments of this invention.
[0020] FIGS. **31-33** show a parallel keeper used alone, according to embodiments of this invention.
DETAILED DESCRIPTION OF THE INVENTION
[0021] FIGS. **1-6** show an exemplary vise **20**, including a stationary jaw **22** and a movable jaw **24**, and further including several components of this invention in combination. The components can be used separately as disclosed in later figures, or in various combinations. The components can be integrally connected to the vise, such as at manufacturing, but are desirably separate components that can be applied to various vise designs, past, present, and future.
[0022] FIGS. **1-6** show a pair of setup jaw components **30** and **32**, or simply setup jaws, which can be integral to the vise or attached to existing vise jaws **22** and/or **24**. The setup jaws **30** and **32** include connection points for one or more of the vise components of this invention.
[0023] In embodiments of this invention, at least one workpiece lock element is attached to a top surface of at least one of the setup jaws. As shown in FIGS. **1-6**, positive lock **40** is attached to setup jaw **30**. Positive lock **40** is moveable along the top surface of setup jaw **30**, and lockable in a desired place along the top surface. The positive lock **40** preferably includes one or more locking elements, such a threaded element that tightens against the setup jaw **30**. As shown in FIG. **2**, the setup jaw **30** preferably includes a row of detents **35** along a back surface to receive a locking element of the positive lock **40**.
[0024] In embodiments of this invention, at least one workpiece side extension element is attached to a side or end surface of at least one of the jaws. As shown in FIGS. **1-6**, an adjustable side extension **50** is attached, such as by threaded connection, to sides of the setup jaw **30**. The side extension **50** is adjustable in the longitudinal direction of the vise **20** via fastener **52** in elongated opening **54**. Adjustable lock extensions **56** contact the sides of the workpiece to laterally secure the workpiece. In the illustrated embodiment, the lock extensions **56** have two differently sized ends. The extensions **56** are reversible and have a narrow end that can be used for securing small/thin parts (e.g., less than 0.25″ wide), and a wider end for larger parts.
[0025] In embodiments of this invention, a parallel clip **60** is attached to each of the opposing sides of at least one of the setup jaws **30**, **32**. As shown, a parallel clip **60** is attached to both opposing sides of both setup (fixed) jaw **30**, and movable jaw **32**, by a threaded fastener. The parallel clips **60**

hold any added vise parallels **65** to the setup jaws **30**, **32** by a spring force.

[0026] In embodiments of this invention, a parallel keeper **70** is disposed between the parallels **65**. The parallel keeper **70** is formed of a first keeper element **72** including a 'female' detent **76** to receive a 'male' end **78** of a second keeper element **74**. The two keeper elements **72**, **74** fit together to avoid lateral displacement upon tightening the vise jaws **30**, **32**. The parallel keepers **72** and **74** can be used by alone to hold the parallels in place, or in combination with parallel keepers **60** if an application requires additional parallel holding.

[0027] FIGS. **7-16** illustrate positive lock **40** according to one embodiment of this invention. FIG. **7** shows the positive lock **40** attached to a top surface of setup jaw **30**. Positive lock **40** is moveable along the top surface of setup jaw **30**, and lockable in a desired place along the top surface. The setup jaw **30** includes a row of detents **35** along a back surface to receive a locking element of the positive lock **40**. In this embodiment, the positive lock **40** includes threaded elements that tighten against spherical elements that fit into the detents **35**. Referring to FIG. **9**, the positive lock **40** includes a central, spring loaded ball **41** that moves back and forth into and out of detents **35** against spring **42** as the positive lock **40** is moved along the setup jaw **30**. This assists in correct placement of the positive lock **40**. When the desired position is obtained, additional locking ball elements **43** are locked into corresponding, adjacent detents **35** by tightening their corresponding threaded elements **44**. In the illustrated embodiments, the threaded elements **44** are disposed at an angle, and tighten against an intermediary ball **45**, which pushes against ball **43** to secure it in a corresponding detent **35**.

[0028] FIGS. **7-16** further illustrate a pair of, preferably lateral threaded, contact elements **48**, which extend from the side of the positive lock **40** to contact the workpiece or other item between the vise.

[0029] FIG. **17** shows an alternative embodiments, where the threaded elements **44** are designed with an end **47** to fit directly into a detent **35**. FIGS. **18-21** show another embodiment, where one or more conical elements **49** are tightened from a top or bottom to push the locking ball **43** into the detent **35**.

[0030] FIGS. **22** and **23** show opposing adjustable side extensions **50** attached to sides of the setup jaw **30**.

[0031] FIGS. **24-25** show parallel clips **60** attached to each of the opposing sides of the setup jaws **30**, **32**, and holding vise parallels **65** to the setup jaws **30**, **32** by a spring force.

[0032] FIGS. **26-30** show a parallel holder or parallel keeper **70** disposed between the parallels, used in combination with the parallel clips **60**. FIGS. **31-33** show a parallel keeper **70** used alone. The parallel keeper **70** is shown as supported on a second pair of parallels **65′** disposed on an outer side of the parallel clips. The parallel keeper **70** is formed of a first keeper element **72** including a 'female' detent **76** to receive a 'male' end **78** of a second keeper element **74**. The two keeper elements **72**, **74** fit together to avoid lateral displacement upon tightening the vise jaws **30**, **32**. The parallel keeper **70** can be produced in different height sizes, and generally one will want to use a size smaller than the parallel itself.

[0033] The invention illustratively disclosed herein suitably may be practiced in the absence of any element, part, step, component, or ingredient which is not specifically disclosed herein.

[0034] While in the foregoing detailed description this invention has been described in relation to certain preferred embodiments thereof, and many details have been set forth for purposes of illustration, it will be apparent to those skilled in the art that the invention is susceptible to additional embodiments and that certain of the details described herein can be varied considerably without departing from the basic principles of the invention.

Claims

1. An improvement for a workpiece holder, the holder including a pair of jaws, the improvement comprising a lock component connected to one of the pair of jaws.

2. The improvement of claim 1, wherein the lock component is moveable and lockable along a top surface of one of the pair of jaws.

3. The improvement of claim 2, wherein the lock component comprises a threaded element, a ball element, and/or a spring loaded ball.

4. The improvement of claim 3, further comprising a row of detents disposed in the one of the pair of jaws, wherein each of the detents is configured to receive the threaded element, the ball element, and/or the spring loaded ball.

5. The improvement of claim 1, further comprising an adjustable side extension for laterally securing workpieces in the holder.

6. The improvement of claim 1, wherein the lock component comprises a pair of clips, each attached to a side of one of the pair of jaws and together configured to secure a parallel against the one of the pair of jaws.

7. The improvement of claim 1, further comprising a parallel keeper including a first keeper element including a female detent and a second keeper element including a male end that fits within the female detent to reduce lateral displacement upon tightening the jaws.